

a particular tip at, say, room temperature, to get back to the initial temperature. The better the thermal recovery, the faster the operator can work.

Tip-to-ground potential. Tip-to-ground potential is also called 'millivolt leakage.' This is the voltage existing between the soldering iron tip and the workstation common point ground. It should not exceed 2 mV as defined by various standards.

Tip-to-ground resistance. This is the resistance between the tip of the soldering iron and the workstation common point ground. It should be 2 ohms to 5 ohms.

Insulation resistance. The resistance between the soldering iron tip and its heating element is known as insulation resistance. Also called 'isolation resistance,' it is actually a measurement of the isolation of the tip from the secondary winding of the transformer.

Anti-static protection. For soldering static-sensitive components, you need an anti-static soldering station.

Advanced and expensive soldering iron stations use static-dissipative materials in their construction to ensure that static does not build up on the iron itself and damage the components when soldered. Such soldering stations are tagged as 'ESD safe.'

The regular irons will not be ESD-safe but these can perform reasonably well in most cases if you take the usual antistatic precautions when handling components. The tip needs to be well earthed (grounded) in such a case.

Selecting the right tip

Selecting the right tip, depending on the type of the project, is of utmost importance. The right tip will get you ef-

ficient and good-quality results. There are several types of tips available. The most common and widely used tips are mentioned below:

Chisel tips. Chisel tips are one of the most widely used tips. For most soldering jobs, a chisel tip will work



the best. It provides you a tapered edge that can fit between most components with a flat tip that facilitates heat transfer.

Conical tips. This type of tip is cone shaped. It is perfect if your project has hard to reach places.



Conical tips should be used only when there is need to get into tight areas. These do not have as much contact area with the component as chisel tips. They require larger contact time with the component to solder the joint.

Pyramid tips. Pyramid tips too are very useful for small components that



are mounted at hard to reach places. These have the benefit of a flat side, which makes heat transfer faster and

easier. Pyramid tips are a very good alternative to conical tips.

Bevel/single flat tips. Bevel tips give the advantages of both chisel and conical tips. A bevel tip is made with a flat edge together with a round edge



on the tip. You can use the side suited to your project. While selecting such a tip, take care that both sides are of correct size for your project. Having the round and flat parts together can save money.

Blade tips. Blade tips are wider than most tips. These tips are highly



useful for rework and pad levelling applications.

Selecting the right tip size

There are plenty of sizes available for each type of tip. Carefully select the tip that easily fits between components on the board. Do check the components that you want to solder and choose a solder tip of correct size and shape. If your tip is too narrow, it will take much longer for the component to heat up and may damage the component. An extra wide tip may be less efficient in heat transfer and may resist heat transfer. ●

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